

REDLINE / RL-MB-01

CONCEPT STUDY - PROVISIONAL DIMENSIONS - DO NOT BUILD WITHOUT REVIEW

80

52

01 BOARD RL-MB-01, COMPONENT SIDE / 2:1

Thickness 1 - notches R4.2 at the four screws - top face at Z = 12

ANTENNA KEEP-OUT (NO PARTS, NO COPPER)

LAYOUT RULES

- ANT1 at the board edge, keep-out clear of copper and parts (checked).
- ANT2 faces the polymer lid; nothing metallic above it.
- U2 near the centre of the board, held by four rigid points.
- BT1 under the board: check the antenna tuning with it in place.
- J1 and SW1 line up with the openings in the -X wall (checked).

GENERAL NOTES

- Dimensions in mm. Third-angle projection. General tolerances to be defined (ISO 2768-m proposed).
- Board outlines: maker or distributor data cited in hardware/netlist.json; others are indicative envelopes.
- Heights, connectors and positions of openings are indicative and must be measured on the chosen parts.
- Geometry and sheets generated by cad/generate.py (build123d 0.13.0); checks run with --check.

BOARD RL-MB-01 AND TAMPER MECHANISM

Redline tracker and black box, board RL-MB-01 (nRF9151)

DRAWING NO.

RL-HW-02

SCALE

2:1 board, 3:1 detail

REV

0

SHEET

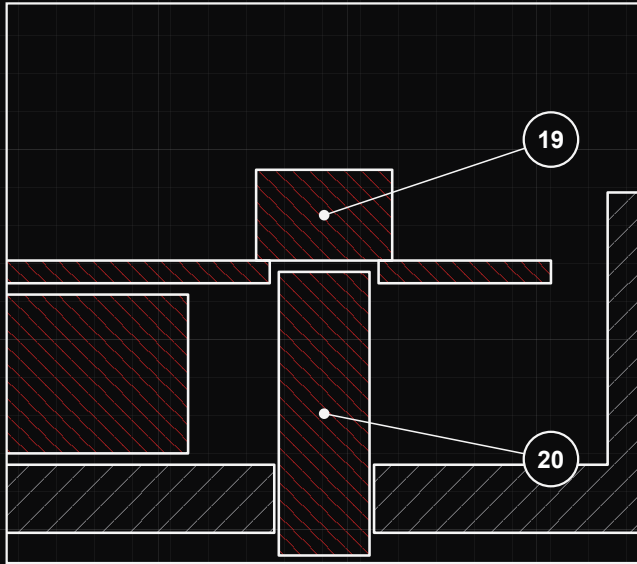
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DRAWN BY: M. ABAAQIL

DATE: 2026-10-05

MATERIAL: to be defined

REV	DESCRIPTION	DATE
0	Hardware concept RL-MB-01	2026-10-05

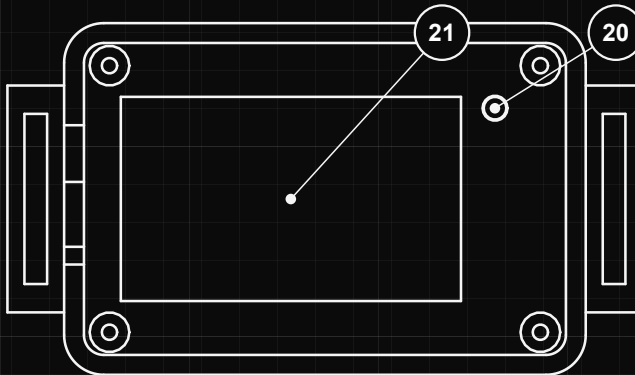


DETAIL B: TAMPER PLUNGER, SECTION Y = +16 / 3:1

Mounting surface pushes PL1 up: TS1 stays open. Remove the device: spring drops PL1, TS1 closes.



LID, INSIDE FACE / 0.75:1: D3.4 hole for the light pipe, no metal over ANT2



BASE WITH BT1 ON FOAM AND PL1 (BOARD REMOVED) / 0.75:1